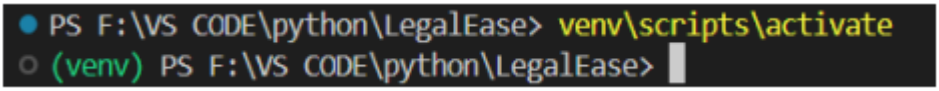
MILESTONE 5: Deployment.

In Milestone 5, the focus is on deploying the LegalEase application locally using FastAPI as the backend and Streamlit as the frontend. This milestone involves setting up a virtual environment, configuring dependencies, and verifying that all modules—from the Gemini API integration to document generation and downloads—work seamlessly in a real-world local environment. This ensures your system is production-ready and paves the way for cloud deployment in future stages.

Activity 5.1: Preparing the Application for Local Deployment.

Step 1: Set Up a Virtual Environment.

To ensure environment isolation and consistent dependency management, create and activate a virtual environment:



In VS Code terminal:

python -m venv venv

Activate it on Windows:

venv\Scripts\activate

After activation, you should see something like:

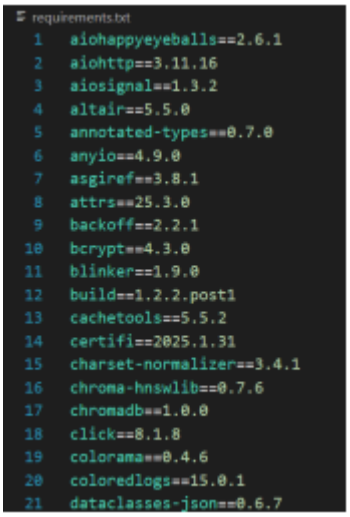
(venv) C:\LegalEase>

This means the virtual environment is active.

Step 2: Install Dependencies.

Ensure that the backend (FastAPI) and frontend (Streamlit) dependencies are installed from a requirements.txt:

pip install -r requirements.txt



The required libraries include:

fastapi

uvicorn

streamlit

python-docx

fpdf

Pillow

requests

google-generativeai

python-dotenv

Step 3: Configure Environment Variables.

Create a .env file to securely store your Gemini API key, rather than putting the key directly into your Python source code. Your documentation specifically instructs you to create this file for the Gemini API key.

For example:

GEMINI_API_KEY=your_api_key_here

Python application can then read this value using python-dotenv.

Project structure:

```
LegalEase/
|
├── venv/
├── .env
├── requirements.txt
|
├── main.py
├── routes.py
├── app.py
|
├── ai_core/
|   └── gemini_generator.py
|
├── utils/
|   ├── docx_generator.py
|   ├── pdf_generator.py
|   └── ...
|
└── assets/
    └── logo.png
```

Activity 5.2: Testing and Verifying Local Deployment.

Step 1: Launch the FastAPI Backend

Start the backend server using Uvicorn:

uvicorn main:app --reload

```
(venv) PS F:\VS CODE\python\LegalEase> uvicorn legalEaseAPI.main:app --reload
INFO:     Will watch for changes in these directories: ['F:\\VS CODE\\python\\Leg
alEase']
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)INFO:
  Started reloader process [15168] using WatchFiles
INFO:     Started server process [17360]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
```

This starts the backend server locally.

Step 2: Launch the Streamlit Frontend.

Run the app.py file to start the LegalEase UI:

```
(venv) PS F:\VS CODE\python\LegalEase> streamlit run frontend/app.py

  You can now view your Streamlit app in your browser.

  Local URL: http://localhost:8501
  Network URL: http://192.168.1.3:8501
```

Open another terminal and run the Streamlit application:

streamlit run app.py

Step 3: Test All Functionalities.

1. Enter the required details

- Document Type
- Parties involved
- Terms & Conditions
- Effective Date

2. Click "Generate Document"

The application sends the information to the Gemini AI and generates the legal document.

3. Preview the document

Check whether the generated document appears correctly in the preview.

4. Edit the document

Use the editing option to modify the generated text if necessary.

5.Check for:

o Proper layout with logo and footer

o Clean formatting of bullet points

o Correct embedding of terms in the document

6. Download the document

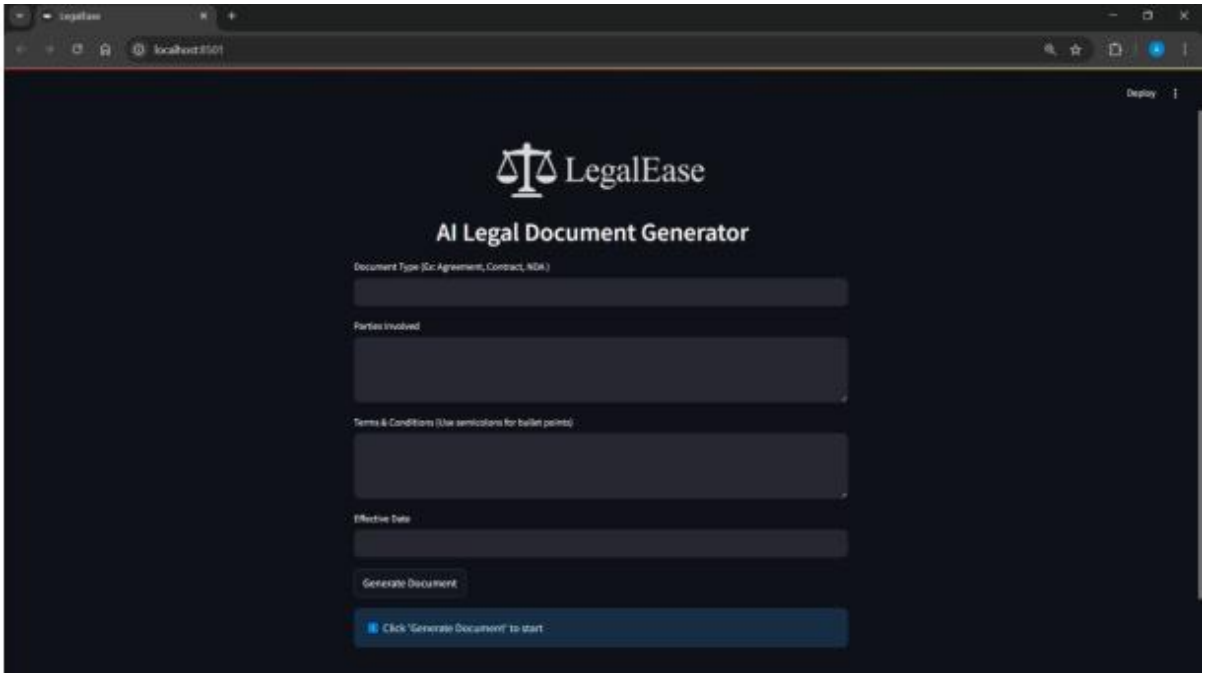
Test all three formats:

.txt

.docx

.pdf

Exploring LegalEase:



1. Document Type

Purpose: Type of legal document to generate.

Examples:

- Agreement
- Contract
- NDA (Non-Disclosure Agreement)
- Lease Agreement
- Employment Offer Letter

Example input: Freelance Work Contract.

2. Parties Involved

Purpose: Names and roles of the individuals or entities involved in the document.

Examples:

- John Doe (Freelancer), ABC Corp (Client)
- Alice Smith (Tenant), XYZ Realty (Landlord)

Example input: Jane Doe (Service Provider), TechNova Inc. (Client).

3. Terms & Conditions (Use semicolons for bullet points)

Purpose: Specific clauses or rules that define the agreement. Separate each bullet point

using a semicolon.

Examples:

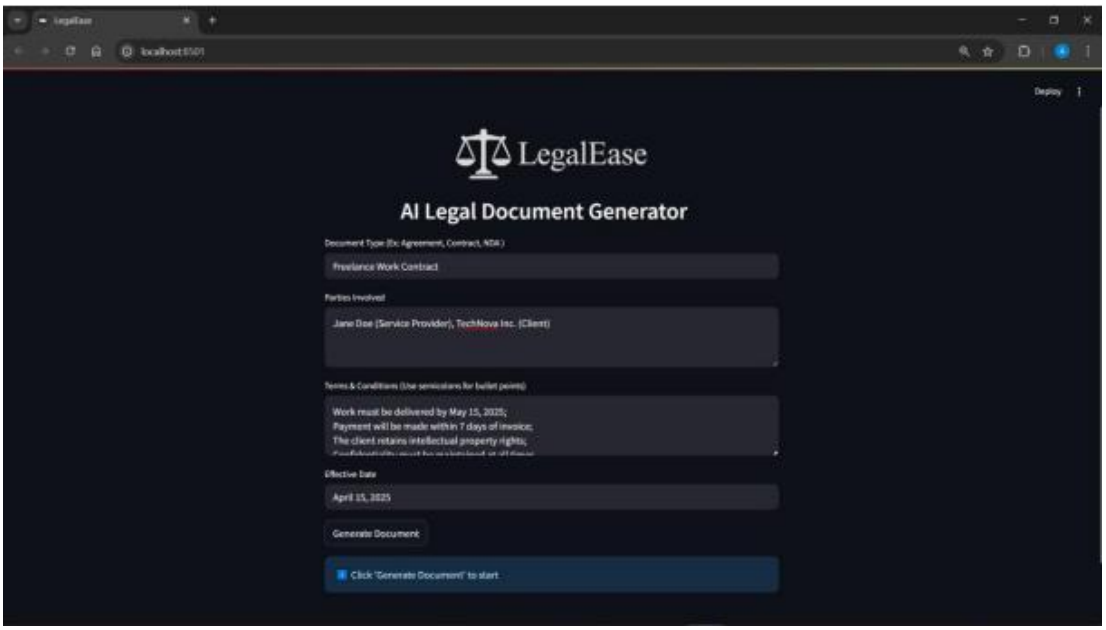
- Payment to be made within 30 days of invoice;
- The provider agrees to deliver work by the agreed deadline;
- Confidentiality must be maintained at all times;
- Either party may terminate with 15 days notice

4. Effective Date

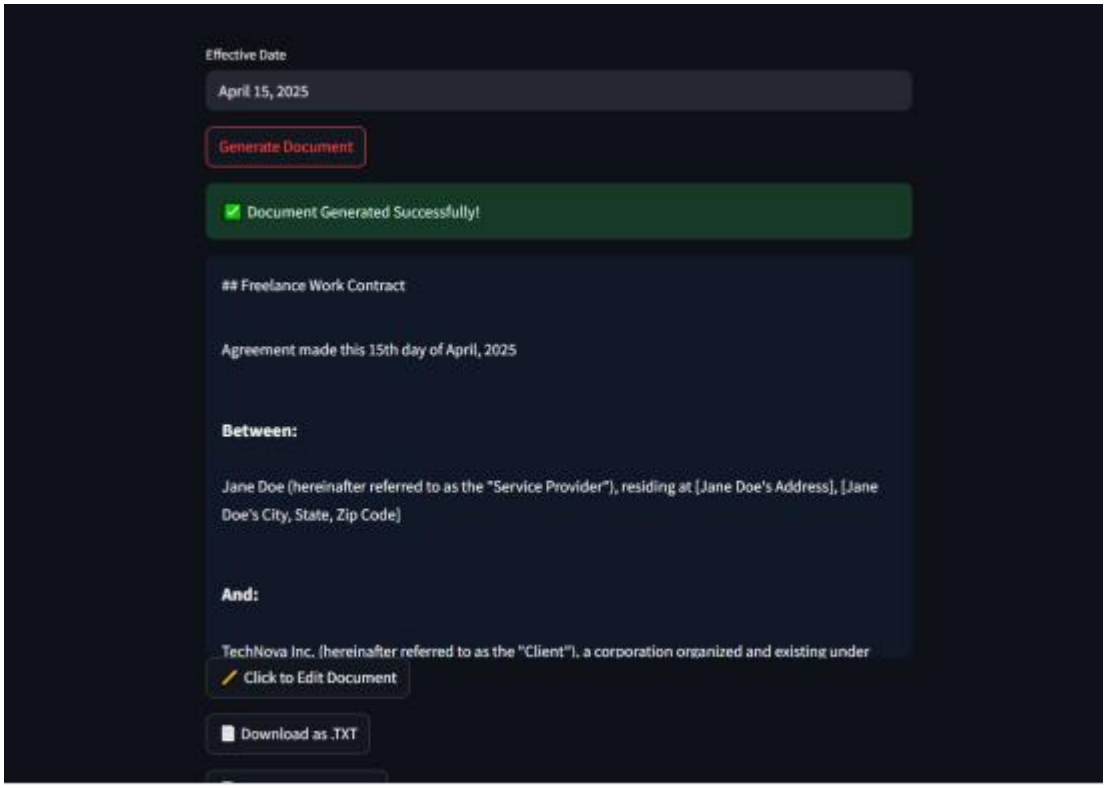
Purpose: The date when the agreement becomes legally valid.

Examples:

- April 10, 2025
- 10/04/2025
- April 15, 2025

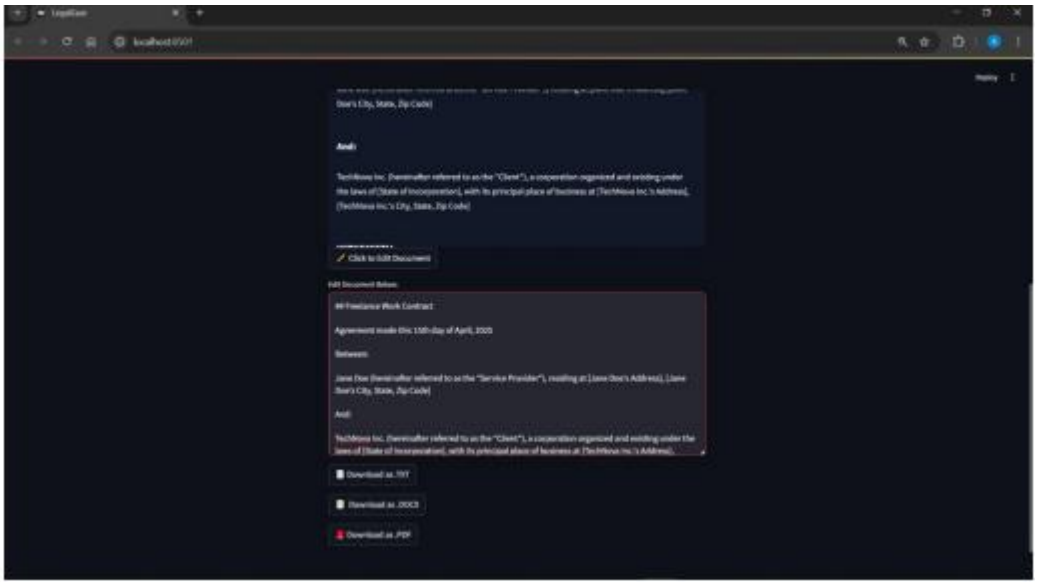


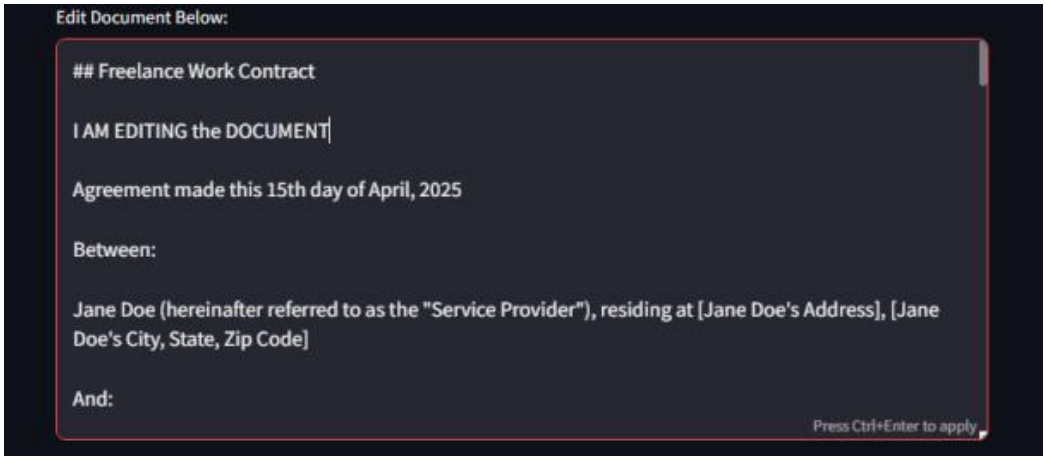
Document Generated



Editing the Document

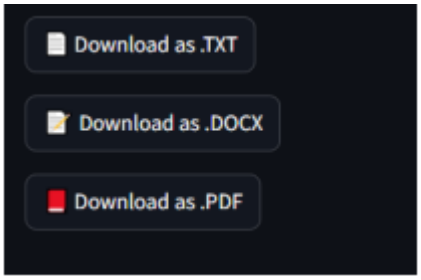
If the user wants to edit the document, they can do so by clicking on the " Click to Edit Document" Botton. A new window opens up where the document can be edited.



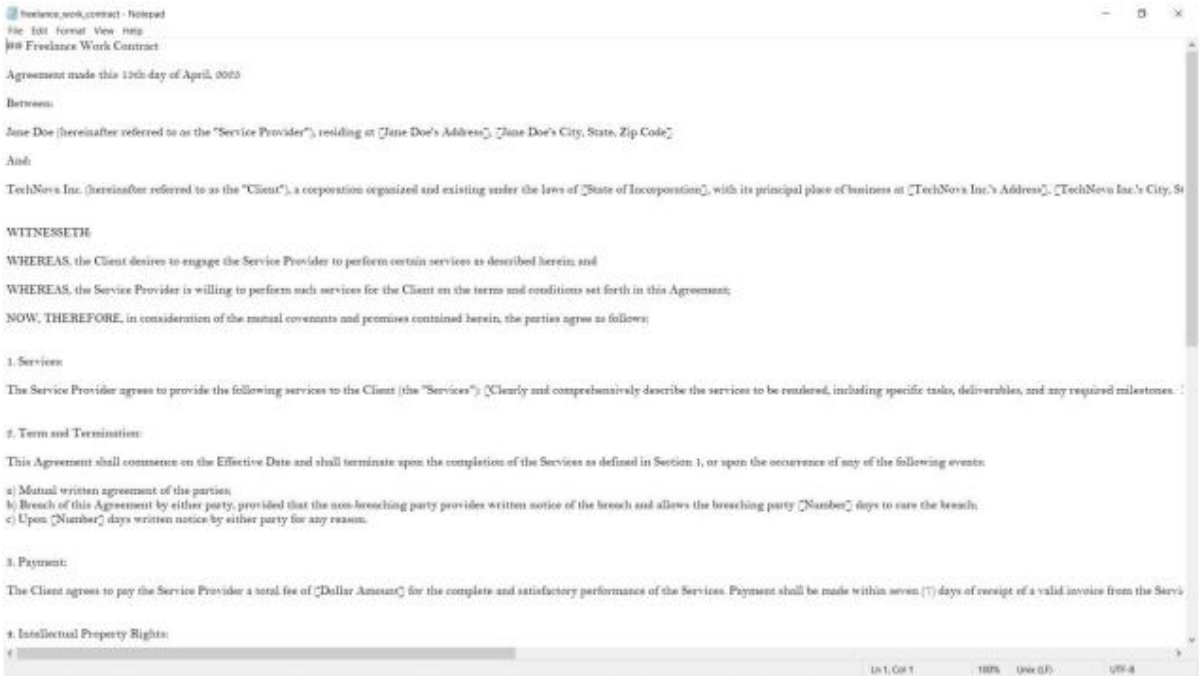


Downloading/ Saving the Generated Document.

We can Download and save the documents in any of the following formats.



.txt


freelance_work_contract - Notepad

File Edit Format View Help

Freelance Work Contract

Agreement made this 12th day of April, 2025

Between:

Jane Doe (hereinafter referred to as the "Service Provider"), residing at [Jane Doe's Address], [Jane Doe's City, State, Zip Code]

And:

TechNova Inc. (hereinafter referred to as the "Client"), a corporation organized and existing under the laws of [State of Incorporation], with its principal place of business at [TechNova Inc.'s Address], [TechNova Inc.'s City, St

WITNESSETH:

WHEREAS, the Client desires to engage the Service Provider to perform certain services as described herein; and

WHEREAS, the Service Provider is willing to perform such services for the Client on the terms and conditions set forth in this Agreement;

NOW, THEREFORE, in consideration of the mutual covenants and promises contained herein, the parties agree as follows:

1. Services:

The Service Provider agrees to provide the following services to the Client (the "Services"): [Clearly and comprehensively describe the services to be rendered, including specific tasks, deliverables, and any required milestones.

2. Term and Termination:

This Agreement shall commence on the Effective Date and shall terminate upon the completion of the Services as defined in Section 1, or upon the occurrence of any of the following events:

a) Mutual written agreement of the parties;
b) Breach of this Agreement by either party, provided that the non-breaching party provides written notice of the breach and allows the breaching party [Number] days to cure the breach;
c) Upon [Number] days written notice by either party for any reason.

3. Payment:

The Client agrees to pay the Service Provider a total fee of [Dollar Amount] for the complete and satisfactory performance of the Services. Payment shall be made within seven (7) days of receipt of a valid invoice from the Servi

4. Intellectual Property Rights:

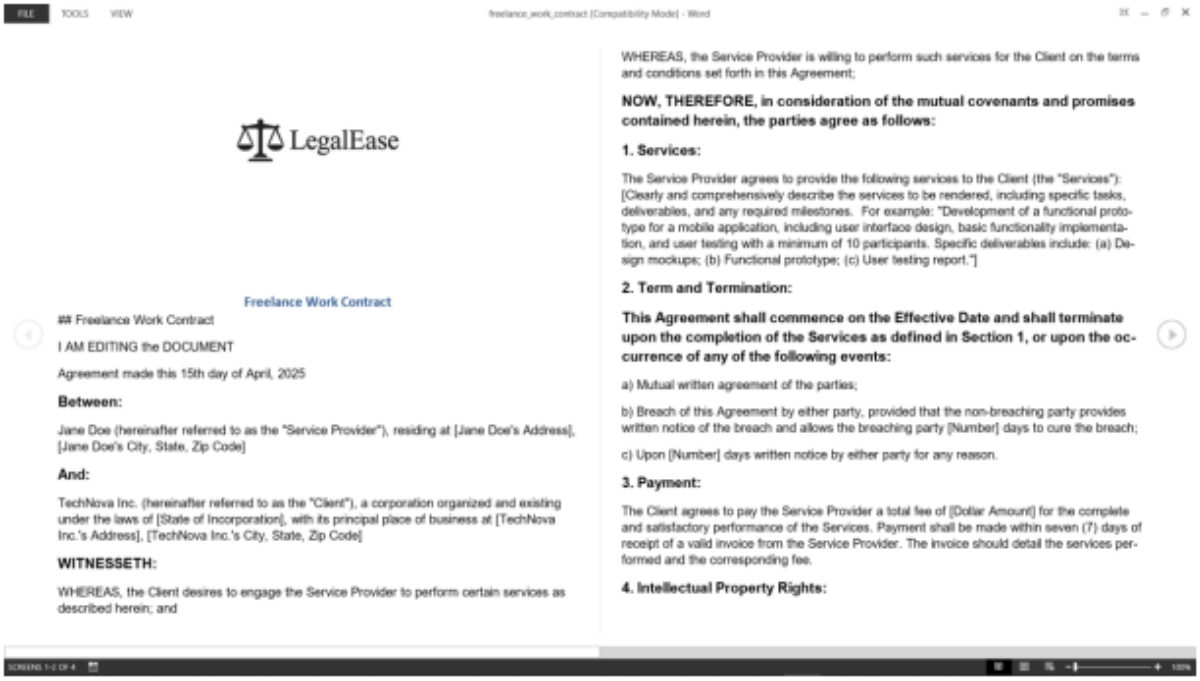
Ln 1, Col 1 100% Unix (LF) UTF-8

.docx

The document provided inculcates the use of the logo (front page) as well as the footer in the last page.

SMARTBRIDGE
Let's Bridge the Gap

Smart Internz



FILE TOOLS VIEW freelance_work_contract [Compatibility Mode] - Word

LegalEase

Freelance Work Contract

Freelance Work Contract

I AM EDITING the DOCUMENT

Agreement made this 15th day of April, 2025

Between:

Jane Doe (hereinafter referred to as the "Service Provider"), residing at [Jane Doe's Address], [Jane Doe's City, State, Zip Code]

And:

TechNova Inc. (hereinafter referred to as the "Client"), a corporation organized and existing under the laws of [State of Incorporation], with its principal place of business at [TechNova Inc.'s Address], [TechNova Inc.'s City, State, Zip Code]

WITNESSETH:

WHEREAS, the Client desires to engage the Service Provider to perform certain services as described herein; and

WHEREAS, the Service Provider is willing to perform such services for the Client on the terms and conditions set forth in this Agreement;

NOW, THEREFORE, in consideration of the mutual covenants and promises contained herein, the parties agree as follows:

1. Services:

The Service Provider agrees to provide the following services to the Client (the "Services"): [Clearly and comprehensively describe the services to be rendered, including specific tasks, deliverables, and any required milestones. For example: "Development of a functional prototype for a mobile application, including user interface design, basic functionality implementation, and user testing with a minimum of 10 participants. Specific deliverables include: (a) Design mockups; (b) Functional prototype; (c) User testing report."]

2. Term and Termination:

This Agreement shall commence on the Effective Date and shall terminate upon the completion of the Services as defined in Section 1, or upon the occurrence of any of the following events:

a) Mutual written agreement of the parties;

b) Breach of this Agreement by either party, provided that the non-breaching party provides written notice of the breach and allows the breaching party [Number] days to cure the breach;

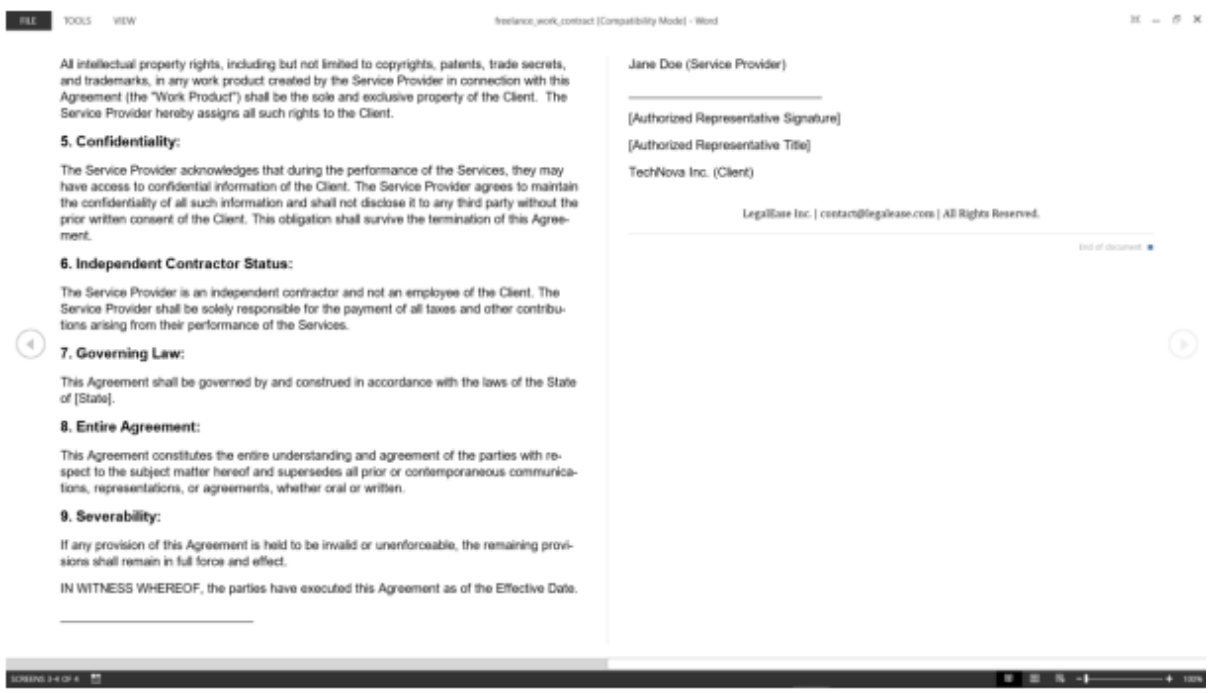
c) Upon [Number] days written notice by either party for any reason.

3. Payment:

The Client agrees to pay the Service Provider a total fee of [Dollar Amount] for the complete and satisfactory performance of the Services. Payment shall be made within seven (7) days of receipt of a valid invoice from the Service Provider. The invoice should detail the services performed and the corresponding fee.

4. Intellectual Property Rights:

SCREENS 1-2 OF 4 100%



FILE TOOLS VIEW freelance_work_contract [Compatibility Mode] - Word

All intellectual property rights, including but not limited to copyrights, patents, trade secrets, and trademarks, in any work product created by the Service Provider in connection with this Agreement (the "Work Product") shall be the sole and exclusive property of the Client. The Service Provider hereby assigns all such rights to the Client.

5. Confidentiality:

The Service Provider acknowledges that during the performance of the Services, they may have access to confidential information of the Client. The Service Provider agrees to maintain the confidentiality of all such information and shall not disclose it to any third party without the prior written consent of the Client. This obligation shall survive the termination of this Agreement.

6. Independent Contractor Status:

The Service Provider is an independent contractor and not an employee of the Client. The Service Provider shall be solely responsible for the payment of all taxes and other contributions arising from their performance of the Services.

7. Governing Law:

This Agreement shall be governed by and construed in accordance with the laws of the State of [State].

8. Entire Agreement:

This Agreement constitutes the entire understanding and agreement of the parties with respect to the subject matter hereof and supersedes all prior or contemporaneous communications, representations, or agreements, whether oral or written.

9. Severability:

If any provision of this Agreement is held to be invalid or unenforceable, the remaining provisions shall remain in full force and effect.

IN WITNESS WHEREOF, the parties have executed this Agreement as of the Effective Date.

_____

Jane Doe (Service Provider)

_____

[Authorized Representative Signature]

[Authorized Representative Title]

TechNova Inc. (Client)

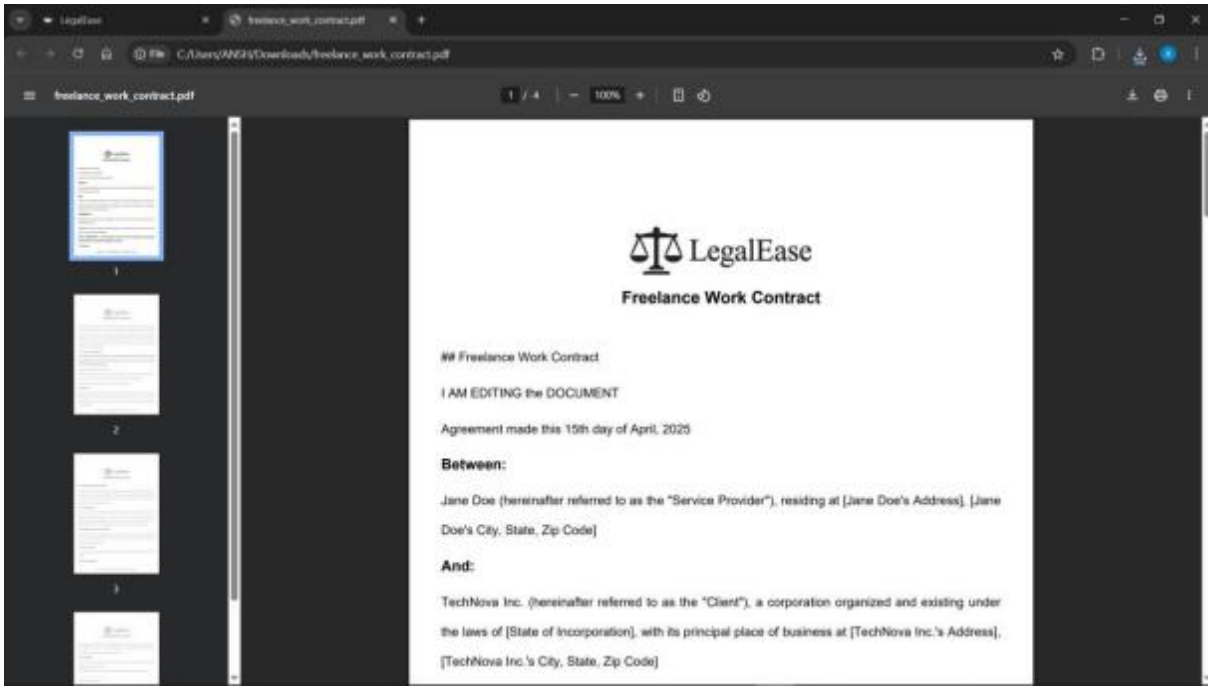
LegalEase Inc. | contact@legalease.com | All Rights Reserved.

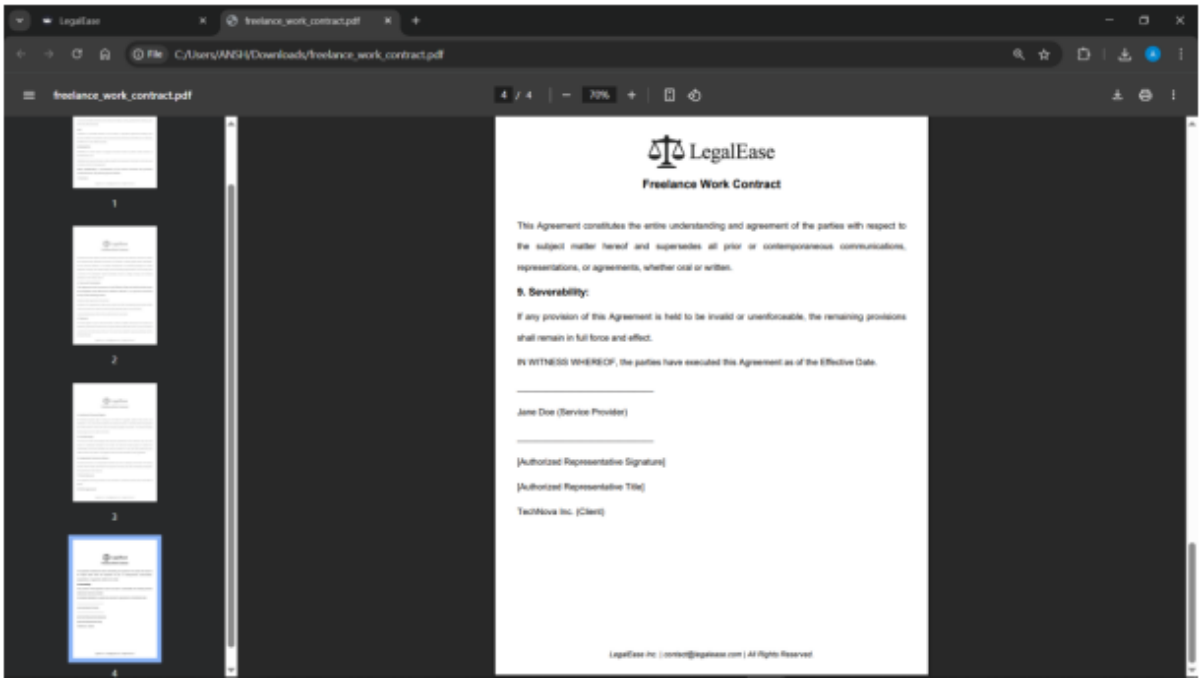
End of document

SCREENS 3-4 OF 4 100%

.pdf

This format provided the user with a well formatted and ready to use legal document with Logo and Footer on all the pages.





Conclusion:

LegalEase represents a transformative step in bridging the accessibility gap in legal services through the power of artificial intelligence. This project demonstrates how emerging technologies can simplify complex legal language, providing users—regardless of legal background—with a clear, concise understanding of legal documents. By integrating advanced natural language processing (NLP) models with user-centric design, LegalEase empowers

individuals to navigate legal content with confidence, thereby promoting inclusivity and informed decision-making. The platform's ability to analyze and simplify legal text, highlight key clauses, and provide plain-language summaries enhances comprehension and reduces dependency on costly legal consultations for basic understanding. This democratization of legal knowledge is critical in a world where legal literacy remains low, especially among marginalized populations.

Furthermore, LegalEase supports multilingual capabilities, ensuring broader reach and utility across different regions and demographics. From a technical standpoint, LegalEase showcases efficient integration of backend AI models, including document parsing, summarization algorithms, and user-friendly web interfaces. The project's architecture is designed to ensure scalability, data privacy, and real-time responsiveness—key elements for any modern SaaS solution. Through iterative development, testing, and feedback, the system has been refined for optimal performance and user satisfaction.

In conclusion, LegalEase is more than a technological tool—it is a catalyst for legal empowerment. By fostering transparency, accessibility, and understanding, it upholds the principle that legal information should be a public good, not a privileged resource. Future enhancements could include deeper contract analysis, integration with legal databases, and personalized recommendations. With continued innovation and support, LegalEase has the potential to revolutionize how people interact with the law, making justice more approachable

for all.